

Programming Techniques – Using C Language

Pointers and Arrays

Objective:

The purpose of this lab is to get an idea about pointers, arrays and their relationship.

- 1) Type and run the given code segments to understand the usage of pointers in arrays.
 - a) Modify **Line A** of above program to **ip = &ar[3];**
Line B to **(ip)--;** and analyze the output.
- 2) Write and execute the following program and get an idea about relationship between arrays and the pointers.

```
#include <stdio.h>
int main() {
    float ar[4] = {19.6,-7.6,7.65,33.7};
    float *ip ; /* a pointer to a float */
    int k;
    ip = &ar[0]; //Line A
    for (k=0; k<4; k++) {
        printf("values ... %f\n",*ip);
        printf(" addresses %p\n",ip);
        (ip)++; //Line B
    }
}
```

```
#include <stdio.h>
int main (){
    int numbers[5];
    int * p;
    int n;
    p = numbers;
    *p = 10;
    p++;
    *p = 20;
    p = &numbers[2];
    *p = 30;
    p = numbers + 3;
    *p = 40;
    p = numbers;
    *(p+4) = 50;
    for (n=0; n<5; n++)
        printf("Array Values %d = %d\n",n,numbers[n]);
    return 0;
}
```

- 3) Learn how we can print out an array either using the array notation or by dereferencing a pointer. The following code illustrates this:

```
#include <stdio.h>
int my_array[] = {1,23,17,4,-5,100};
int *ptr;
int main(void) {
    int i;
    ptr = &my_array[0]; /* <-- Line A --point the pointer to the first element of the array */
    printf("\n\n");
    for (i = 0; i < 6; i++){
        printf("my_array[%d] = %d ",i,my_array[i]); /*<-- Line B */
        printf("ptr + %d = %d\n",i, *(ptr + i)); /*<-- Line C */
    }
    return 0;
}
```

- Change the “Line C” as follows and run the program.
printf("ptr + %d = %d\n",i, *ptr++);
- Change the “Line C” as follows and run the program
printf("ptr + %d = %d\n",i, *(++ptr));
- Change the “Line A” as follows and run the program
ptr = my_array;
- Change the “Line A” as follows and run the program
my_array = ptr;

4)

- a) Create two integer type arrays with size 10.
int array1[10], array2[10];
- b) Fill the first array with suitable integers.
- c) Copy the content of **array1** to **array2** using pointers, get the help from given code segment. Check the result.

```
int *ip1, *ip2 = &array2[0];
int *ep = &array1[10];
for(ip1 = &array1[0]; ip1 < ep; ip1++)
    *ip2++ = *ip1;
```

5) Consider the two rows of elements.

1, 3, 5, 7, 9

2, 4, 6, 8, 10

a) Implement these in C using pointers.

i) Put them into two arrays called **A** and **B**.

ii) Get another array called **C** from the **A** and **B** arrays like,

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

6) Type the given uncompleted code which helps to understand the strings and pointers.

```
#include <stdio.h>

char strA[40] = "For testing purpos";
char strB[40];

int main(void) {
    char *pA; /* a pointer to type character */
    char *pB; /* another pointer to type character */
    puts(strA); /* show string A */
    pA = strA; /* point pA at string A */

    pB = strB; /* point pB at string B */
    printf("\n");

    while(*pA != ' ') /* line A */
    {
        ; /* line B */
    }
    *pB = ' '; /* line C */
    printf("%s\n",strB); /* show strB on screen */
    return 0;
}
```

The above code should copy strA to strB.

a) Correct the Line A and C.

b) Write the code segment in Line B.